

Lab 4 Worksheet - Practice

- add something about loading a csv and having to use # Split the one column into four `energy_raw <- separate( energy_raw, col = 1, into = c("State", "2021", "2022", "2023"), sep = "," )`
- make sure str extract is clear `str_extract(.x, "\d+\.?\d*")`

Q1. Extract and clean text patterns from strings using regular expressions.

You are given a messy dataset of energy sources with inconsistent units and spacing.

```
energy_sources <- c("Solar (MWh)", "Wind (GWh)", "Coal (MWh)", "Hydro (kWh)")
```

- a. Use `str_detect()` to identify all energy sources that contain the substring "Wind" or "Hydro" (hint: use the `|` operator in regex).
- b. Extract the unit type (e.g., MWh, GWh, kWh) using `str_extract()`. Make sure your pattern handles extra spaces or hyphens.
- c. Clean the names to keep only the energy source name, trimming whitespace and removing parentheses.
- d. Create a clean tibble called `energy_clean` with two columns: source and unit.
- e. Discussion: Why might regex cleaning be important before performing joins or aggregations?

Q2. Joining State Datasets: Socioeconomic + Regional Data

We'll use built-in datasets to explore US state-level data. `state.x77` contains metrics such as population, income, and life expectancy. `state.region` classifies each state into one of four regions: Northeast, South, North Central, West.

```
# Load datasets  
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
# Convert matrix to data frame
states <- data.frame(state.x77)

# Add state names as a new column
states$state <- rownames(states)

# select and reorder columns
states <- states |>
  select(state, Income, Population, Illiteracy)
# table for regions
regions <- data.frame(state = state.name, region = state.region)
```

- Perform a left join to combine socioeconomic and regional data.
- Perform an inner join. How many rows remain, and why is it the same or different?
- Create a summary table that shows average Income and average Life Exp by region.
- Perform an anti join to check if any state names don't match between datasets. What might cause mismatches like this in real projects?
- Discussion: Why would you prefer a left join over an inner join when merging state-level datasets?
- Use `pivot_longer()` to make the socioeconomic data longer — convert the numeric columns into key-value pairs.

Q3. Mapping

Let's visualize data by state.

```
library(ggplot2)
library(maps)
library(dplyr)

us_map <- map_data("state")

state_data <- states |>
```

```
left_join(regions, by = "state") |>
mutate(state = tolower(state))
```

Use a join to combine the `us_map` with the `state_data`

```
us_map_joined <- ...
```

Create a choropleth map:

```
ggplot(us_map_joined, aes(long, lat, group = group, fill = Income)) +
  geom_polygon(color = "white") +
  coord_fixed(1.3) +
  theme_minimal() +
  scale_fill_viridis_c() +
  labs(title = "Average Income by State")
```

- b. Swap Income for Life Exp to visualize life expectancy instead. Try changing the color scale using `scale_fill_gradient()` or `scale_fill_gradient2()`

Q4. Quarto Dashboard Setup

Create a new file `ev-dashboard.qmd` and add:

```
---
title: "EV Power Dashboard"
format: dashboard
---

Add some sections:

### Overview

Your project introduction.

### Map Visualization

# Paste your ggplot map code here

### Data Table

# Show your joined dataset

head(us_map_joined)
```

Then render:

```
quarto preview
```